

Class 10 NCERT Maths

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Chapter 1: Real Numbers (Core Basics Edition)

CHAPTER 1: REAL NUMBERS

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

Core basics (RED) ko miss mat karna!

IS PDF MEIN RED RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

Ye woh cheezein hain jo tujhe pehle se aani chahiye thi.

Inhe DHYAAN se padh - inke bina chapter samajh nahi aayega.

Agle page pe SAARE core basics ek saath diye hain.

Phir asli Chapter 1 shuru hota hai.

CORE BASICS - MISS MAT KARNA (RED PAGE)

Ye page sirf un cheezon ka hai jo Class 7-8-9 mein padhayi jaati hain. 10th ka Chapter 1 inhi pe khada hai.

1) FACTOR KYA HOTA HAI?

Factor = jo number kisi doosre number ko POORA (bina remainder) divide kar de.

Example: 12 ke factors = 1, 2, 3, 4, 6, 12
(kyunki ye sab 12 ko poora divide karte hain)

2) MULTIPLE KYA HOTA HAI?

Multiple = number ki "table" ke numbers.

Example: 3 ke multiples = 3, 6, 9, 12, 15, ...
(Factor chhota hota hai, multiple bada.)

3) PRIME, COMPOSITE, AUR 1

PRIME = sirf 2 factors (1 aur khud). Jaise 2,3,5,7,11

COMPOSITE = 2 se zyada factors. Jaise 4,6,8,9,10

1 NA prime hai NA composite (sirf 1 factor hai).

2 ekloti EVEN prime number hai.

4) POWER / EXPONENT (chhota upar wala number)

2^3 ka matlab 2 ko 3 baar multiply = $2 \times 2 \times 2 = 8$.

Yahan 2 = base, 3 = power/exponent.

a^2 ko "a square", a^3 ko "a cube" bolte hain.

5) SQUARE ROOT (sqrt ya the root symbol)

Square root = power ka ULTA.

$\sqrt{9} = 3$ kyunki $3 \times 3 = 9$.

$\sqrt{25} = 5$ kyunki $5 \times 5 = 25$.

Symbol: tick-jaisa nishaan (root sign) number ke upar.

Is PDF mein hum " $\sqrt{n}$ " likhte hain (jaise $\sqrt{2}$).

6) p/q MEIN q ZERO KYUN NAHI HO SAKTA?

p/q ka matlab p ko q se DIVIDE karna.

Kisi cheez ko 0 se divide karna ALLOWED nahi

($5/0$ ka koi answer exist nahi karta - undefined).

Isliye rational number p/q mein hamesha $q \neq 0$.

7) HCF (Highest Common Factor)

HCF = do numbers ka sabse BADA common factor.

12 ke factors: 1,2,3,4,6,12

18 ke factors: 1,2,3,6,9,18

Common: 1,2,3,6 -> sabse bada = 6 -> HCF = 6

8) LCM (Least Common Multiple)

LCM = do numbers ka sabse CHHOTA common multiple.

12 ke multiples: 12,24,36,48...

18 ke multiples: 18,36,54...

Common sabse chhota = 36 -> LCM = 36

9) NUMBER TYPES (jaldi recap)

Natural (N): 1,2,3,... Whole (W): 0,1,2,3,...

Integers (Z): ...-2,-1,0,1,2...

Rational (Q): p/q form ($q \neq 0$), jaise $1/2$, 0.25

Irrational: p/q mein nahi aata, jaise $\sqrt{2}$, π

Real (R): rational + irrational dono.

10) TERMINATING vs NON-TERMINATING DECIMAL

Terminating = decimal khatam ho jaye: 0.5, 0.25

Non-terminating repeating = pattern repeat: 0.333...

Non-terminating non-repeating = na khatam na pattern:

$\sqrt{2} = 1.41421356...$ (ye IRRATIONAL hota hai)

11) CO-PRIME NUMBERS

Co-prime = do numbers jinka HCF = 1 (1 ke alawa koi common factor nahi).

Example: 8 aur 15 -> common factor sirf 1 -> co-prime.

(Dono ka khud prime hona zaroori nahi - 8,15 dono composite hain par phir bhi co-prime hain.)

NOW: ASLI CHAPTER 1 SHURU

TOPIC 1: FUNDAMENTAL THEOREM OF ARITHMETIC

STATEMENT:

"Har composite number ko prime numbers ke product (multiplication) ke roop mein likha ja sakta hai, aur ye tareeka UNIQUE hota hai (order chhod ke)."

[CORE] "Product of primes" matlab prime numbers ko multiply karna. Pehle prime/factor wala page dekh.

EXAMPLE 1: 156 ka prime factorisation

$$\begin{aligned}156 / 2 &= 78 \\78 / 2 &= 39 \\39 / 3 &= 13 \\13 / 13 &= 1\end{aligned}$$

$$156 = 2 \times 2 \times 3 \times 13 = 2^2 \times 3 \times 13$$

[CORE] 2^2 ka matlab 2×2 (power wala page yaad kar).

EXAMPLE 2: 3825 ka prime factorisation

$$\begin{aligned}3825 / 3 &= 1275 \\1275 / 3 &= 425 \\425 / 5 &= 85 \\85 / 5 &= 17 \\17 / 17 &= 1\end{aligned}$$

$$3825 = 3^2 \times 5^2 \times 17$$

TOPIC 2: HCF AND LCM BY PRIME FACTORISATION

METHOD:

HCF = common prime factors ka product (LOWEST power)
LCM = saare prime factors ka product (HIGHEST power)

[CORE] HCF/LCM ka basic matlab RED page pe diya hai (point 7 aur 8). Pehle wo samajh, phir ye method.

EXAMPLE 3: 96 aur 404 ka HCF aur LCM

$$96 = 2^5 \times 3$$
$$404 = 2^2 \times 101$$

$$\text{HCF} = 2^2 = 4 \quad (\text{common prime 2, lowest power})$$
$$\text{LCM} = 2^5 \times 3 \times 101 = 9696 \quad (\text{all primes, highest power})$$

Verify: $\text{HCF} \times \text{LCM} = 4 \times 9696 = 38784$
 $96 \times 404 = 38784 \quad (\text{match!})$

[CORE] Sirf 2 numbers ke liye: $\text{HCF} \times \text{LCM} = \text{product}$ of the two numbers. (3 numbers pe ye rule nahi chalta.)

EXAMPLE 4: Ek diya ho toh doosra nikaalo

Do numbers ka HCF = 9, LCM = 90, ek number = 18.
Doosra number = ?

$$\text{HCF} \times \text{LCM} = \text{number1} \times \text{number2}$$
$$9 \times 90 = 18 \times \text{number2}$$
$$810 = 18 \times \text{number2}$$
$$\text{number2} = 810 / 18 = 45$$

TOPIC 2B: HCF / LCM WORD PROBLEMS (EXAM IMPORTANT)

PEHCHAAN (kaunsa use karna hai):

- "Ek saath", "together", "same time" -> LCM
- "Sabse bada", "maximum", "largest" -> HCF

EXAMPLE 5: Bells (ek saath bajna) -> LCM

Teen ghantiyan 6, 12 aur 18 second pe bajti hain.
Kab dobara ek saath bajengi?

-> LCM of 6, 12, 18

$$6 = 2 \times 3$$
$$12 = 2^2 \times 3$$
$$18 = 2 \times 3^2$$
$$\text{LCM} = 2^2 \times 3^2 = 36$$

Answer: 36 second baad ek saath bajengi.

EXAMPLE 6: Sabse badi tape (exact measure) -> HCF

Do rassiyan 18 m aur 24 m hain. Sabse badi tape
jo dono ko exact (poora) naap sake?

-> HCF of 18, 24

$$18 = 2 \times 3^2$$
$$24 = 2^3 \times 3$$
$$\text{HCF} = 2 \times 3 = 6$$

Answer: 6 m ki tape.

TOPIC 3: IRRATIONAL NUMBERS (sqrt2 is irrational)

[CORE] sqrt(2) ka matlab "woh number jisko square karne pe 2 aaye". Root symbol RED page point 5 mein.

THEOREM USED:

Agar prime p , a^2 ko divide karta hai, toh p , a ko bhi divide karta hai.

PROOF: sqrt2 is irrational (Proof by Contradiction)

Step 1: Maan le sqrt2 rational hai. Toh $\sqrt{2} = p/q$ where $q \neq 0$ and $HCF(p,q) = 1$ (simplest form).

[CORE] $q \neq 0$ kyun? RED page point 6 dekh.
 $HCF(p,q)=1$ matlab fraction simplest form mein.

Step 2: Square dono side:

$$2 = p^2 / q^2 \rightarrow 2 q^2 = p^2 \quad \dots(i)$$

Step 3: 2 divides $p^2 \rightarrow$ 2 divides p .

Step 4: So $p = 2c$.

Step 5: (i) mein daal: $2 q^2 = 4 c^2 \rightarrow q^2 = 2 c^2$

Step 6: 2 divides $q^2 \rightarrow$ 2 divides q .

Step 7: Ab p aur q dono ko 2 divide karta hai, par humne kaha tha $HCF = 1$. CONTRADICTION!

CONCLUSION: Assumption galat. sqrt2 IRRATIONAL hai.

EXAMPLE 7: sqrt3 bhi irrational hai (same method)

1. Maan le $\sqrt{3} = p/q$ ($HCF=1$, $q \neq 0$)

2. $3 q^2 = p^2 \rightarrow$ 3 divides $p^2 \rightarrow$ 3 divides p

3. $p = 3c \rightarrow 3 q^2 = 9 c^2 \rightarrow q^2 = 3 c^2$
 $\rightarrow$ 3 divides q

4. p, q dono ko 3 divide kar raha $\rightarrow HCF=1$ toot gaya

5. Contradiction $\rightarrow$ sqrt3 IRRATIONAL hai.

QUICK RESULTS (yaad rakh):

- $\sqrt{p}$ irrational hota hai jab p prime ho.
- Rational + Irrational = Irrational
- (non-zero) Rational x Irrational = Irrational

Q AND A TIME

(Pehle khud try kar, phir mujhe bhej - main check karunga)

- Q1. 140 ko prime factors ke product mein likho.
- Q2. 26 aur 91 ka HCF aur LCM nikaal (prime factorisation se).
- Q3. $\sqrt{5}$ ko irrational prove kar ($\sqrt{2}$ wala method copy kar).
- Q4. CORE CHECK: 1 prime hai ya composite? Kyun?
- Q5. CORE CHECK: $\frac{5}{0}$ ka answer kya hai? $\frac{p}{q}$ mein q ke baare mein kya rule hai?
- Q6. $3 + 2 \times \sqrt{5}$ ko irrational dikhao.
- Q7. WORD PROBLEM: Do ghantiyan 8 aur 12 minute pe bajti hain. Kab dobara ek saath bajengi? (HCF ya LCM use hoga?)
- Q8. CORE CHECK: 14 aur 25 co-prime hain ya nahi? Kyun?

SUMMARY

1. Fundamental Theorem: har composite number = unique product of primes.
2. HCF = common primes x lowest power.
LCM = all primes x highest power.
3. 2 numbers ke liye: $\text{HCF} \times \text{LCM} = \text{product of numbers}$.
4. $\sqrt{p}$ irrational hai jab p prime ho.
5. Rational + Irrational = Irrational.

CORE BASICS (RED page) ek baar aur revise kar lena - factor, multiple, prime, power, root, $q! \neq 0$, HCF, LCM. Inke bina ye chapter adhoora rahega.